ANALYSTLAB AFRICA

Machine Learning Internship Programme

Official Weekly Assignment

Week 3: Machine Learning Model Development & Performance Evaluation

Document Control

Programme	Machine Learning Internship
Week	Week 3
Project	Machine Learning Model Development & Performance Evaluation
Version	1.0
Issue Date	__________
Submission Deadline	End of Week 3 (Sunday, 11:59 PM WAT)

Business Scenario

Following your business understanding, dataset inspection, and data preparation completed in Weeks 1 and 2, your organization is ready to begin predictive modelling.

As a Junior Machine Learning Engineer, you have been tasked with developing and evaluating machine learning models that can accurately predict business outcomes based on historical data.

Your objective is to build multiple supervised machine learning models, compare their performance, interpret the results, and recommend the most suitable model for deployment.

Learning Objectives

By the end of this assignment, you should be able to:

- Build supervised machine learning models.
- Split datasets into training and testing sets.
- Train multiple classification models.
- Evaluate model performance using standard metrics.
- Compare algorithms objectively.
- Interpret model results from both technical and business perspectives.
- Recommend the most appropriate model for deployment.

Project Options

Continue working on the **same dataset selected in Week 2**.

Business Questions

Your modelling should answer the following:

Customer Churn

1. Which model predicts churn most accurately?
2. Which customer features contribute most to churn prediction?
3. How well can churn be predicted before customers leave?
4. Which evaluation metric is most appropriate for this business problem?
5. Which model would you recommend for deployment?

Assignment Tasks

Part 1 – Data Preparation

Using your Week 2 dataset:

- Review cleaned data
- Verify encoded variables
- Verify feature engineering
- Handle any remaining imbalance
- Prepare the final modelling dataset

Part 2 – Train/Test Split

Split the dataset into:

- Training Set
- Testing Set

Explain:

- Train/Test ratio
- Why this split was selected

Part 3 – Model Development

Develop **at least four classification models**.

Recommended algorithms:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- Gradient Boosting
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naïve Bayes

Choose at least **four**.

Part 4 – Model Evaluation

Evaluate every model using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

Compare model performance using tables and visualizations.

Part 5 – Feature Importance

Identify the most important predictors.

Include:

- Feature Importance Chart
- Discussion
- Business interpretation

Part 6 – Model Comparison

Compare all developed models.

Discuss:

- Advantages
- Limitations
- Computational complexity
- Interpretability
- Business suitability

Recommend the best-performing model with clear justification.

Part 7 – Business Report

Prepare a **3–5 page report** covering:

- Modelling approach
- Algorithms selected
- Evaluation results
- Model comparison
- Feature importance
- Business recommendations
- Next steps

Minimum Project Requirements

Machine Learning Models

Your project **must** include:

- At least **4 classification models**
- Training and testing datasets
- Performance comparison table

Model Evaluation

Include:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

Visualizations

Include at least **10 visualizations**, such as:

- Confusion Matrices
- ROC Curves
- Precision-Recall Curves
- Feature Importance Charts
- Model Comparison Charts
- Learning Curves
- Prediction Distribution

Documentation

Include:

- Well-commented notebook
- Markdown explanations
- Business interpretations
- Professional formatting

GitHub Repository

Your repository should include:

- README.md
- Jupyter Notebook
- Business Report
- Model Evaluation Report
- Requirements.txt
- Processed Dataset

Deliverables

Submit the following:

1. Jupyter Notebook (.ipynb)
2. Business Report (PDF or Word)
3. Model Evaluation Report
4. Feature Importance Report
5. Processed Dataset (.csv)
6. README.md
7. GitHub Repository
8. Google Drive Folder (Anyone with the link can view)
9. Submit using the official Week 3 Submission Form

Assessment Criteria

Criteria	Weight
Business Understanding & Problem Framing	10%
Data Preparation	15%
Model Development	25%
Model Evaluation & Comparison	25%
Business Interpretation & Recommendation	15%
Documentation & Professionalism	10%
Total	**100%**

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 3 project.
- Publish a LinkedIn post describing your machine learning workflow, the algorithms you compared, and the business insights gained.
- Share your best-performing model, explain why it outperformed the others, and summarize your evaluation results.
- Publish an X (Twitter) post summarizing your project.
- Tag **AnalystLab Africa** on LinkedIn.
- Tag **@analystlabafric** on X.
- Include the hashtag **#AnalystLabAfrica** on both platforms.

Professional Branding Reminder

Building a machine learning model is only part of the job. Professional Machine Learning Engineers evaluate multiple algorithms, justify their model choices using objective metrics, and communicate technical findings in a way that supports business decisions. This assignment is designed to simulate that real-world workflow, helping you build both technical expertise and a strong portfolio.

Submission Checklist

☐ Jupyter Notebook (.ipynb)

☐ Business Insights Report

☐ Statistical Analysis Report

☐ Feature Engineering Documentation

☐ Updated Data Dictionary

☐ Final Machine Learning Dataset (.csv)

☐ README.md Updated

☐ GitHub Repository Updated

☐ LinkedIn Post Published

☐ X (Twitter) Post Published

☐ Google Drive Folder Submitted